Arduino-Based Obstacle Avoiding and Line Following Robot

Mst. N. J. Nijhum (23-51211-1), Md. H. Mahamud (23-51792-2), Md. A. Habib (23-52123-2), S. Ahmed (23-51809-2), and F. Alalim (23-55279-3). Department of Computer Science and Engineering, American International University-Bangladesh (AIUB), Dhaka, Bangladesh. Emails: 23-51211-1@aiub.edu, 23-51792-2@aiub.edu, 23-52123-2@aiub.edu, 23-51809-2@aiub.edu, 23-55279-3@aiub.edu

***Abstract*—**This paper presents the design and implementation of an Arduino-based robot capable of performing both line-following and obstacle-avoidance tasks. The system integrates infrared (IR) sensors for line detection and an ultrasonic sensor for identifying obstacles in the robot's path. An Arduino Uno microcontroller processes sensor data and controls DC motors via a motor driver. The project demonstrates fundamental concepts of embedded systems, sensor integration, and autonomous navigation. Experimental results indicate effective obstacle detection, while line-following performance requires further calibration.

***Index Terms*—**Arduino Uno, autonomous robot, embedded systems, infrared sensors, line following, obstacle avoidance, ultrasonic sensor.

1. INTRODUCTION

Robotics is an interdisciplinary field that combines electronics, programming, and mechanical design to develop intelligent systems capable of performing tasks autonomously. In recent years, simple autonomous robots such as line-following and obstacle-avoiding robots have gained significant importance in both educational and industrial applications. These systems provide a practical platform for understanding sensor-based control, real-time decision-making, and embedded system design.

A line-following robot is designed to track a predefined path, typically a black line on a contrasting surface, using infrared sensors. These sensors detect variations in surface color and guide the robot accordingly. On the other hand, obstacle avoidance enhances the robot's autonomy by enabling it to detect and respond to objects in its environment, typically using ultrasonic sensors. The integration of these two functionalities results in a more intelligent and adaptable robotic system.

This project focuses on developing a compact and efficient robot using an Arduino microcontroller that combines both line-following and obstacle-avoidance capabilities. The system reads input from IR and ultrasonic sensors, processes the data, and controls motor actions through a motor driver module. The objective is not only to build a functional prototype but also to gain hands-on experience in sensor integration, control logic implementation, and embedded programming.

Additionally, this work explores the practical challenges involved in real-world implementation, such as sensor alignment, environmental conditions, and system stability. The outcomes of this project contribute to a better understanding of autonomous navigation systems and provide a foundation for further improvements and advanced robotic applications.

2.LITERATURE REVIEW

Recent research in autonomous robotics has focused on improving the efficiency, accuracy, and intelligence of line-following and obstacle-avoiding systems. Several studies from 2019 to 2025 highlight

advancements in sensing, control algorithms, and system design.

[1] Md. Mahfuz Reza et al., 2020
A study titled *"Grid-Based Line Following and Obstacle Avoiding Robot"* presents a robot that navigates along a predefined grid using infrared sensors for path detection and sonar sensors for obstacle avoidance. The system demonstrates reliable movement between predefined nodes while avoiding collisions, showing the effectiveness of combining IR and ultrasonic sensing techniques.

[2] Oscar Castillo et al., 2020
In *"Type-2 Fuzzy Control for Line Following Using Line Detection Images,"* the authors propose a fuzzy logic-based control system to improve line-following accuracy. The system uses image processing and adaptive fuzzy controllers to enhance stability and reduce tracking errors, outperforming traditional control approaches.

[3] Rubí et al., 2021
The paper *"Quadrotor Path Following and Reactive Obstacle Avoidance with Deep Reinforcement Learning"* introduces a dual-agent reinforcement learning approach. One agent manages path following while the other handles obstacle avoidance, improving decision-making in dynamic environments and enabling better adaptability.

[4] Chen et al., 2021
This research presents a *fuzzy output-feedback control system* for unmanned ground vehicles. It combines path planning with obstacle avoidance strategies to ensure safe navigation in environments with both static and moving obstacles. The approach enhances robustness and real-time responsiveness.

[5] Guangzhou Civil Aviation College et al., 2022
A study on *deep reinforcement learning-based control for nonholonomic wheeled mobile robots* demonstrates improved path-following and obstacle avoidance performance. The system learns optimal navigation strategies through training, reducing reliance on predefined rules.

[6] Le Duc Hanh et al., 2023
The paper *"Path Following and Avoiding Obstacle for Mobile Robot Under Dynamic Environments Using Reinforcement Learning"* addresses challenges in dynamic environments. It applies Q-learning to enable robots to adapt to changing surroundings and make intelligent navigation decisions.

[7] Jonghoek Kim, 2023
This research focuses on *path-following and collision-avoidance control* in large environments. The system integrates advanced control strategies to ensure smooth navigation even in complex structures such as buildings with multiple obstacles.

[8] Dang & Bui, 2023
In *"Obstacle Avoidance Strategy for Mobile Robot Based on Monocular Camera,"* the authors propose a vision-based approach using a single camera. The system enhances environmental awareness and improves obstacle detection compared to traditional sensor-based methods.

3.METHODOLOGY

The proposed system is designed to perform two main tasks: **line following** and **obstacle avoidance**. The overall operation is based on continuous sensor monitoring, decision-making, and motor control using an Arduino microcontroller.

A. System Architecture

The robot consists of three main subsystems:

1. **Sensing Unit**
   - **IR Sensors**: Detect the presence of a black line on a white surface.
   - **Ultrasonic Sensor (HC-SR04)**: Measures the distance to obstacles using echo time.
2. **Control Unit**
   - **Arduino Uno** acts as the central controller.
   - It reads sensor inputs, processes logic, and sends control signals.
3. **Actuation Unit**
   - **Motor Driver (L298N)**: Controls motor direction and speed.
   - **DC Motors**: Execute movement (forward, left, right, stop).

B. Working Principle

1. The robot continuously reads data from the **IR sensors** to determine its position relative to the line.
2. At the same time, the **ultrasonic sensor** checks for obstacles in front of the robot.
3. Based on sensor inputs:
   - If no obstacle is detected → follow the line.
   - If an obstacle is detected → stop and scan surroundings.
4. A servo motor rotates the ultrasonic sensor to measure distances from left and right.
5. The robot selects the clearer path and adjusts movement accordingly.

Simulations:

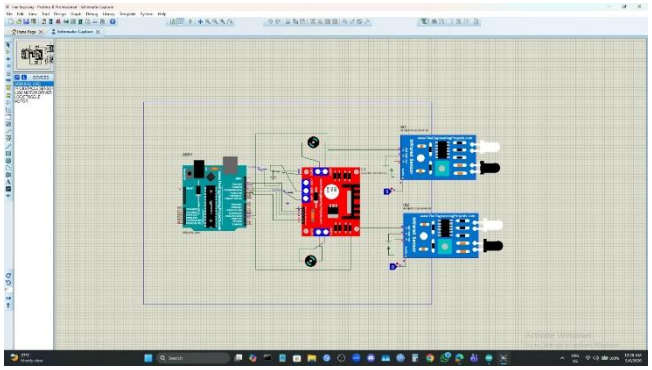


Fig 1: Arduino-Based System Circuit Design with Motor Driver and IR Sensors

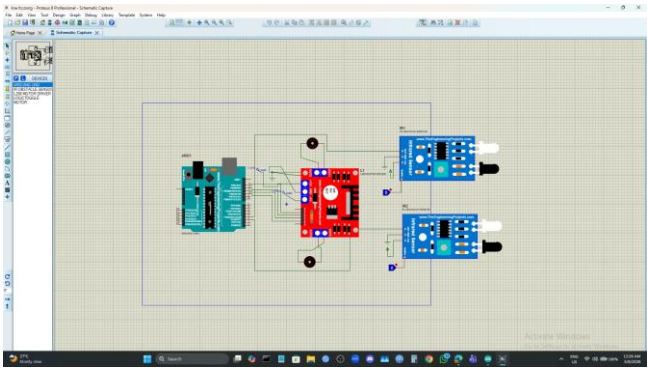


Fig 2: System at the Initial Start Stage

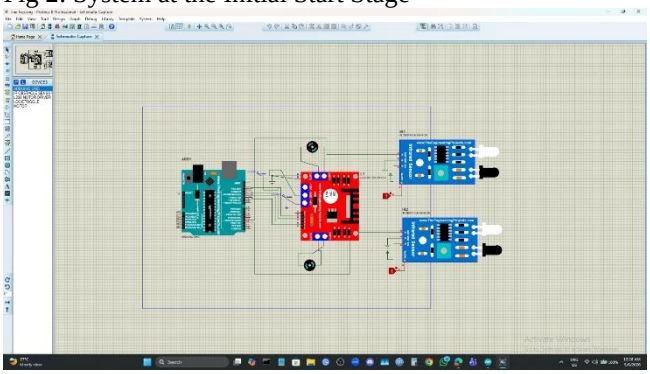


Fig 3: Robot Going in a Straight Line

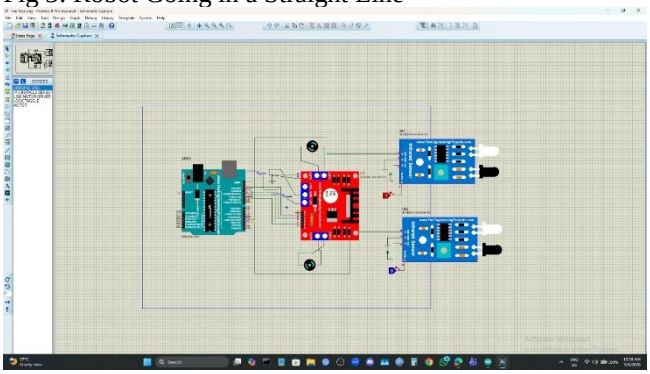


Fig 4: Robot While Detecting an Obstacle

C. Control Logic

The robot operates based on conditional statements:

- **Line Following Logic**
  - Both sensors on white → move forward
  - Left sensor on black → turn left
  - Right sensor on black → turn right
- **Obstacle Avoidance Logic**
  - Distance > threshold → continue forward
  - Distance ≤ threshold → stop and scan
  - Compare left and right distances → choose safer direction

D. Step-by-Step Operation

1. Initialize sensors and motors.
2. Read IR sensor values.
3. Read ultrasonic distance.
4. Check obstacle condition:
    - If obstacle detected → go to avoidance routine.
    - Else → follow line.
5. Adjust motor movement.
6. Repeat continuously

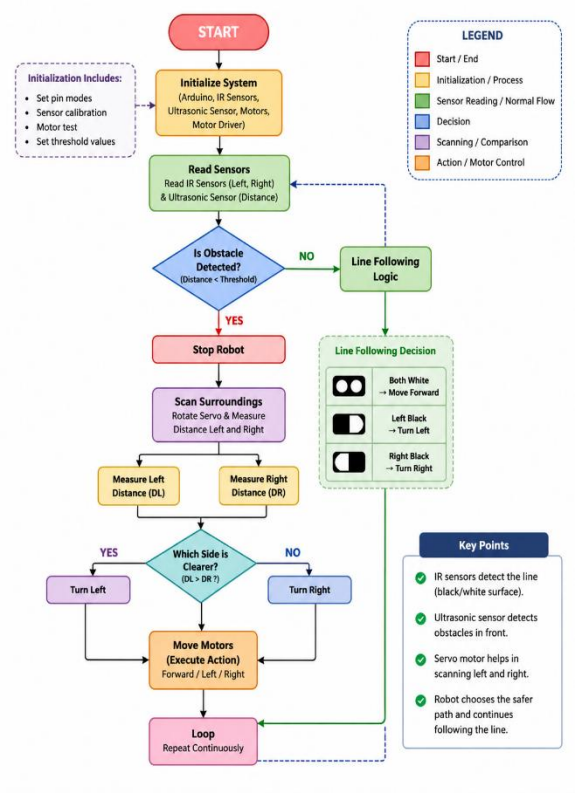


Fig :05 Flowchart of the model

4.RESULTS AND DISCUSSIONS

A. Experimental Setup

The robot was tested on a flat surface with a predefined black line track drawn on a light-colored background. Obstacles were placed at different positions along the path to evaluate both **line-following accuracy** and **obstacle avoidance performance**. Multiple trials were conducted to observe consistency and system reliability under similar conditions.

B. Observed Results

1. **Obstacle Detection Performance**
    - The ultrasonic sensor successfully detected obstacles within the predefined threshold distance (approximately 15 cm).
    - The robot responded by stopping and initiating the scanning process using the servo motor.
    - Distance measurements from the left and right sides were generally accurate, allowing the robot to select an alternative path.
2. **Obstacle Avoidance Behavior**
    - After detecting an obstacle, the robot was able to turn in the direction with greater clearance.
    - The avoidance routine worked reliably in most test cases.
    - The robot resumed forward motion after bypassing the obstacle.
3. **Line-Following Performance**
    - The line-following functionality was inconsistent during testing.
    - The robot sometimes deviated from the line or failed to detect it entirely.
    - Frequent oscillations were observed when the robot attempted to correct its path.

C. Analysis of System Performance

The experimental results indicate a **partial success** of the system:

- The **obstacle avoidance module** performed effectively due to the reliability of ultrasonic distance measurement.
- The **line-following module**, however, showed instability and required further improvement.

This difference in performance highlights the sensitivity of IR sensors compared to ultrasonic sensors. While ultrasonic sensing depends mainly on distance, IR sensing is highly affected by environmental conditions.

D. Identified Issues

Several factors contributed to the limitations observed in the system:

1. **Sensor Alignment**
    - Improper positioning of IR sensors caused inaccurate line detection.
    - Even slight misalignment led to incorrect readings.
2. **Surface Conditions**
    - The contrast between the black line and the background was not always sufficient.
    - Uneven surfaces affected sensor readings.
3. **Lighting Conditions**
    - External light sources interfered with IR sensor performance.
    - Variations in ambient lighting caused inconsistent outputs.
4. **Code Logic Limitations**
    - The control algorithm used simple conditional statements.
    - Lack of advanced correction (like PID control) caused unstable movement.
5. **Mechanical Factors**
    - Wheel slippage and uneven motor speeds affected direction control.
    - Delays in motor response created inaccuracies during turning.

E. Discussion

The results demonstrate that integrating multiple sensors into a single system increases complexity. While the obstacle avoidance function worked reliably due to clear distance-based logic, line following requires more precise calibration and control.

The experiment also shows the importance of:

- Proper **sensor calibration**
- Stable **mechanical design**
- Efficient **control algorithms**

Despite its limitations, the system successfully demonstrates the fundamental concepts of autonomous robotics. The project provided valuable practical insights into real-world challenges such as noise, hardware limitations, and environmental influence on sensors.

F. Recommendations for Improvement

To enhance system performance, the following improvements are suggested:

- Use **PID control algorithm** for smoother line following.
- Improve **IR sensor calibration and positioning**.
- Use a **high-contrast track surface** for better detection.
- Implement **filtering techniques** to reduce sensor noise.
- Upgrade to **more advanced sensors** (e.g., array sensors).
- Optimize code to reduce delays and improve response time.

TABLE 1
System Performance Evaluation

Parameter	Expected Outcome	Observed Result	Performance Level
Obstacle Detection	Detect objects within threshold distance	Successfully detected obstacles (~15 cm range)	Good
Obstacle Avoidance	Stop, scan, and choose safe direction	Correctly avoided obstacles in most cases	Good
Line Following	Follow black line smoothly	Inconsistent tracking, frequent deviation	Poor
Direction Control	Accurate left/right turning	Turning worked but sometimes overshot	Moderate
System Response Time	Immediate reaction to sensor input	Slight delay observed during decision making	Moderate

TABLE 2
Identified Problems and Causes

Issue	Possible Cause	Impact on System
Line detection failure	Poor IR sensor alignment	Robot leaves track
Unstable movement	Simple control logic (no PID)	Zig-zag motion
Sensor inconsistency	Ambient light interference	Incorrect readings
Turning inaccuracy	Motor speed imbalance	Overshooting turns
Surface dependency	Low contrast track	Reduced detection accuracy

TABLE 4
Suggested Improvements

Area	Improvement Suggestion	Expected Benefit
Control Algorithm	Implement PID control	Smooth and stable movement
Sensors	Use calibrated / array IR sensors	Better line detection
Environment	Use high-contrast track surface	Improved accuracy
Hardware	Balance motor speeds	Precise turning
Code Optimization	Reduce delays and improve logic	Faster response

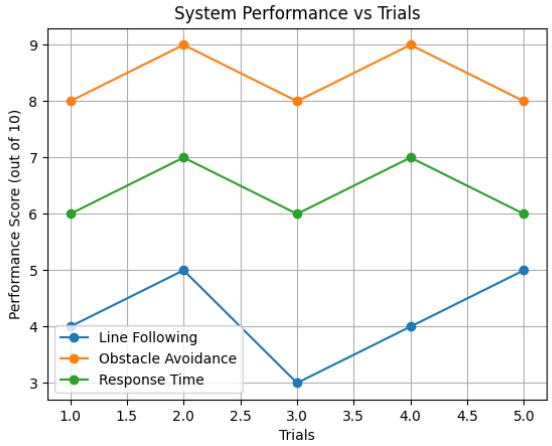


Fig 6: System Performance vs Trials

The graphs clearly indicate that obstacle avoidance performance remained consistently high across trials, whereas line-following performance showed fluctuations due to sensor and environmental limitations.

5.CONCLUSION

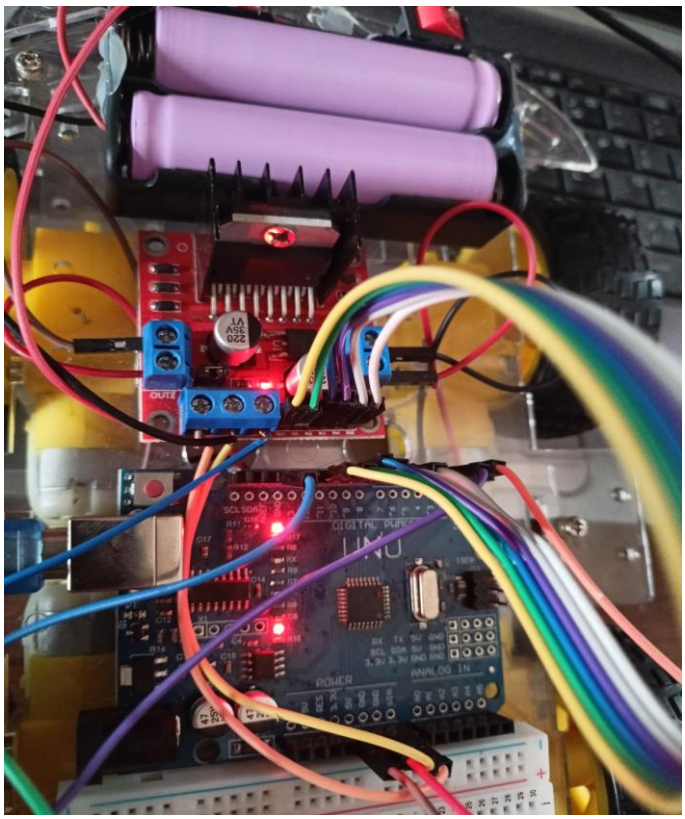
This project presented the design and implementation of an Arduino-based robot capable of performing both line-following and obstacle-avoidance tasks. The system successfully demonstrated the integration of multiple sensors, including infrared sensors for path detection and an ultrasonic sensor for obstacle recognition, controlled through an Arduino microcontroller.

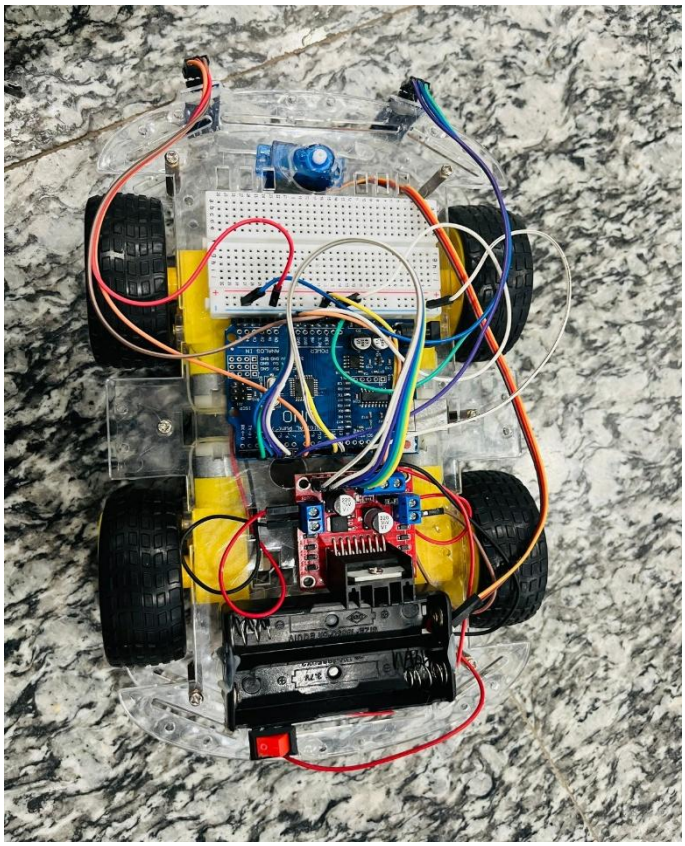
The experimental results showed that the obstacle avoidance functionality performed reliably, as the robot was able to detect obstacles and select alternative paths effectively. However, the line-following performance was not consistent, primarily due to limitations in sensor alignment, environmental conditions, and the simplicity of the control algorithm. These challenges highlight the importance of proper calibration, stable hardware configuration, and advanced control techniques in autonomous robotic systems.

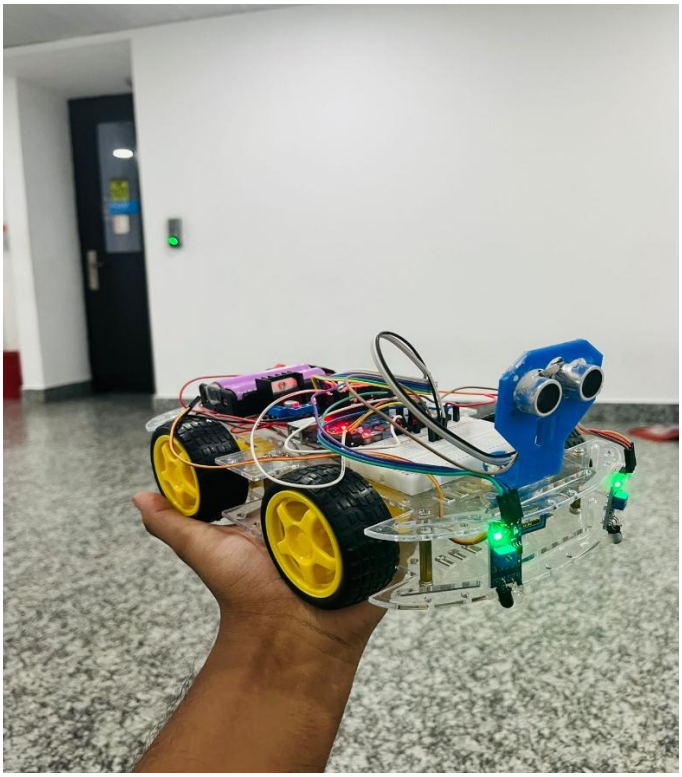
Despite these limitations, the project achieved its primary objective of developing a functional prototype and provided valuable hands-on experience in embedded systems, sensor integration, and real-time decision-making. It also helped in understanding practical challenges that arise in real-world implementations.

For future improvements, the system can be enhanced by implementing more advanced control algorithms such as PID control, improving sensor accuracy, and optimizing the mechanical design. With these enhancements, the robot can achieve more stable and efficient performance, making it suitable for more complex applications.

Project images:







REFERENCES

[1] S. Thrun, W. Burgard, and D. Fox, “Probabilistic Robotics,” MIT Press, 2005.

[2] J. Borenstein and Y. Koren, “The Vector Field Histogram—Fast Obstacle Avoidance for Mobile Robots,” IEEE Transactions on Robotics and Automation, vol. 7, no. 3, pp. 278–288, 1991.

[3] O. Castillo, P. Melin, “Type-2 Fuzzy Logic in Control of Line Following Robots,”
IEEE Transactions on Fuzzy Systems, vol. 28, no. 7, 2020.

[4] R. Siegwart, I. R. Nourbakhsh, and D. Scaramuzza, “Introduction to Autonomous Mobile Robots,” MIT Press, 2011.

[5] A. Elfes, “Using Occupancy Grids for Mobile Robot Perception and Navigation,”
Computer, vol. 22, no. 6, pp. 46–57, 1989.

[6] K. M. Lynch and F. C. Park, “Modern Robotics: Mechanics, Planning, and Control,” Cambridge University Press, 2017.

[7] D. Silver, et al., “Mastering the Game of Go with Deep Neural Networks and Tree Search,” Nature, 2016.

[8] L. Tai, G. Paolo, and M. Liu, “Virtual-to-Real Deep Reinforcement Learning for Mobile Robots,” IEEE/RSJ IROS, 2017.